

Bachelor of Arts in Applied Information Technology program, University of Jamestown

CS:344 PYTHON

Professor: Jessica Schwartz

July 12th, 2026

Project Description

The Smart Paycheck Budget Allocator will help users decide how to divide their take-home pay after taxes and other deductions. The user will enter the amount of each paycheck and their recurring monthly bills. The program will then suggest how much money should go toward essential bills, savings, a rainy-day fund, and a splurge fund.

The intended user is anyone who wants help organizing their money without using a complicated budgeting app. This program can be especially useful for people who receive multiple paychecks during the month and want to make sure their important expenses are covered before spending money on nonessential items.

The program will set aside \$50 from each paycheck for regular savings and aim to place \$200 per month into a rainy-day fund. The splurge fund will have a maximum limit of \$1,200. After showing the suggested budget, the program will ask the user whether the budget needs to be adjusted.

The first section will require the user to enter all income and bill information manually. It will not connect to bank accounts or automatically track purchases. The program will also assume that the user enters valid positive numbers and that the budget covers one month at a time.

Inputs

The program will receive the following information from the user:

- Take-home pay after deductions
- Number of paychecks received during the month
- Recurring bill names
- Recurring bill amounts
- Menu choices
- Whether the user wants to readjust the budget
- Any updated category amounts entered during an adjustment

Outputs

The program will produce:

- Total monthly take-home income
- Total amount of recurring bills
- Suggested savings amount
- Suggested rainy-day fund amount
- Suggested splurge fund amount
- Remaining balance
- A warning if income does not cover required expenses
- A warning if the splurge amount would exceed \$1,200
- A final budget summary

Algorithm Overview

1. **Get user input.**

Ask the user to enter their take-home pay, number of paychecks, and recurring bills.

2. **Calculate monthly income.**

Determine the user's total monthly income based on the information entered.

3. **Allocate the budget.**

Calculate recurring bills, savings, the rainy-day fund, and the suggested splurge fund.

4. **Check the budget.**

Use decisions to determine whether the user is within budget, over budget, or needs to adjust spending.

5. **Display the budget summary.**

Show the user's income, expenses, savings, remaining balance, and budget recommendations.

6. **Ask if the user wants to readjust the budget.**

Use a loop to allow the user to make changes until they choose to finish.

Planned Structure and Functions

get income()

Gets the user's take-home pay and number of paychecks.

Input: Paycheck amount, number of paychecks

Output: Monthly income

calculate budget ()

Calculates the budget totals.

Input: Income and expenses

Output: Budget totals and remaining balance

display summary ()

Displays the completed budget summary.

Input: Budget information

Output: Budget report

adjust budget ()

Allows the user to make changes to the budget.

Input: User's choice and updated amounts

Output: Updated budget